

A Guide to Construct Non-canonical *Wh*-questions: A Cross-linguistic Perspective

[非典疑問句的建構指南：一個跨語言的視角]

Wei-Tien Dylan Tsai [蔡維天]

National Tsing Hua University [國立清華大學]

This paper provides a comprehensive analysis of non-canonical *wh*-questions from a cross-linguistic perspective. This study claims that they are encoded through the interaction of various functional elements, prosodic constraints and pragmatic construals. Our investigation reveals that they combine a *wh*-expression with a modal element that can be either lexical or silent. Prosodically, they are characterized by distinct pitch and stress patterns. In terms of semantics, these constructions involve negation over modal quantification in conjunction with various not-at-issue contents such as expectations and presuppositions. Pragmatically, they change information-seeking into a denial/disapproval act, typically raising objections to the at-issue content within the scope of sentential *wh*-adverbs. Finally, we suggest that the origin of non-canonical *wh*-questions may well trace back to the hierarchical arrangement of causal and source questions: Namely, they are disrupted when the speaker is no longer interested in the cause-effect relationship, holding instead a negative attitude towards the interlocutor's remarks or behavior. Our approach thus sheds new light on the complex nature of non-canonical *wh*-questions in relation to their interrogative counterparts.

Keywords: causality hierarchy, force shift, non-canonical *wh*-questions, prosodic syntax, syntax-pragmatic interface

關鍵詞：使事階級、念力移轉、非典疑問句、韻律語法、句法—語用介面

1. Setting the stage

A typical question seeks information from the addressee, while a non-canonical *wh*-question expresses either illocutionary forces other than interrogation, such as warning and denial, or speaker's attitudes including surprise and annoyance. This paper lays out a few strategies for constructing a non-canonical *wh*-question.

First, we leave out rhetorical questions such as (1) and (2), where the denotation of the *wh*-expression in question is an empty set in the former and a singleton set in the latter (cf. Han 2002, Phan & Tsai 2022):

- (1) 你 說！ 誰 會 做 這種 傷天害理 的 事？！
 Ni shuo! Shei hui zuo zhezhong shangtianhaili de shi?!
 you speak who will do this.CL atrocity DE thing
 'Tell me! Who will do such an atrocity?!'
 (i.e., Nobody will do such an atrocity!)
- (2) 你 說！ 是 誰 把 你 養大 的？！
 Ni shuo! Shi shei ba ni yangda de?!
 you speak be who BA you raise DE
 'Tell me! Who raised you?!'
 (i.e., It's me [the speaker] who raised you!)

In terms of pragmatics, these cases essentially involve making a statement forcefully by asking a question with apparent answers (i.e., *mingzhiguwen* [明知故問] 'knowingly asking a question').

Our argumentation is organized as follows: In Section 2, we address the issue from the morpho-syntactic perspective. It combines a *wh*-expression and a modal (which can be silent), very often accompanied by a focus/mood marker. In Section 3, we show on the prosodic front that non-canonical *wh*-construals are typically associated either with a peculiar intonation, or with a stress shift onto certain functional elements. Section 4 proceeds to investigate the interpretative aspects of non-canonical *wh*-construals. We show that they typically involve negation over modal quantification. In Section 5, we face the challenges posed by the "force shift" phenomena on the pragmatic front. It is proposed that a non-canonical *wh*-question is formed when a cause-effect relationship breaks down, hence shifting its information-seeking function to a non-interrogative speech act. Section 6 concludes this paper.

2. The composition of non-canonical *wh*-questions

As laid out in the seminal work of Hamblin (1973) and Karttunen (1977), a typical *wh*-question essentially combines the speech act of information-seeking with existential quantification over *wh*-expressions (hence a set of propositions as possible/true answers). In the same spirit, we propose that a non-canonical *wh*-expression has two components: one involves negation over modal quantification on the semantic side (see also Cheung 2008), while the other is to substantiate a non-interrogative speech act on the pragmatic side. We therefore define a non-canonical *wh*-question as a constituent question losing its interrogative force to a performative construal of warning, whining, denial, or disapproval, etc.

In this paper, we mainly concern ourselves with the morpho-syntactic aspect of this duality analysis and show how non-canonical *wh*-questions can be built by drawing insights and evidence from typological comparisons. First, consider the following minimal pair to sharpen our intuition:

- (3) a. How did John do this to her?
b. How could John do this to her?!

(3a) is a typical instrumental *wh*-question, whereas (3b) expresses the speaker's disapproval instead of an inquiry for the addressee. It means something like "John did this to her, but it should not be possible (and hence should not be allowed)." In other words, the speaker is in denial of something they believe to be a fact. We would like to suggest that it is this conflict of beliefs that results in the peculiar interpretation of (3b): With appropriate stress/intonation adjustment (cf. Rett & Sturman 2021), the presence of the modal *could* appears to be the key to "shifting" the force from interrogative to disapproval (sometimes characterized as whining, surprise or bewilderment).

Nevertheless, very often we do not find a tensed modal in the Mandarin counterpart of (3b), where a denial/causal *how* merges high in the left periphery, as exemplified by the causal-mirative construal of (4), as well as the disapproval usage of (5) (cf. Lü 1985, Shao 1995, Stepanov & Tsai 2008, Tsai 2008, 2015a, Tang 2011, Pan 2015, Cheng 2021, Tsai & Yang 2022, among many others):^{1,2}

- (4) 阿 Q 怎麼 老是 遲到? (!) (causal-mirative *how*)
Akiu **zenme** laoshi chidao? (!)

Akiu how always late
'How come Akiu was always late?(!)'

- (5) 怎麼 阿 Q 這樣 對 你! (denial-disapproval *how*)
Zenme Akiu zheyang dui ni!
how Akiu this.manner treat you
'How can Akiu treat you like this/as such!'

Interestingly enough, both (4) and (5) can be paraphrased with a lexical modal *hui* 'will'

¹ As an anonymous reviewer points out, *why*-questions may overlap with *how come*-questions in daily usages. Interestingly, Tsai (2008) proposes to sharpen the intuition further by employing a stative predicate: As illustrated by the contrast between the reason question (ia) and the causal question (ib), only the latter sports a mirative (i.e., counter-expectation) construal:

- (i) a. Why is the sky blue?
b. How come is the sky blue?! (It should be cloudy.)

This reason-cause distinction carries over to their Mandarin counterparts, as evidenced by (iia) and (iib):

- (ii) a. 天空 為什麼 是 藍 的?
Tiankong weishenme shi lan de?
sky why be blue DE
'Why is the sky blue?'
b. 天空 怎麼 是 藍 的? !
Tiankong zenme shi lan de?!
sky how.come be blue DE
'How come the sky is blue?! (It should be cloudy.)'

See also the relevant discussion in Section 5.

² Some native speakers might not be able to make a clear distinction in grammaticality between (5) and (i):

- (i) 阿 Q 怎麼 這樣 對 你? !
Akiu zenme zheyang dui ni?!
Akiu how this.manner treat you
'How come Akiu treat you like this/as such?!'

Here the grammatical judgement may be somewhat obscured by the fact that the subject *Akiu* 'Akiu' may further topicalize to the sentence-initial position. So the disambiguation hinges upon the peculiar intonation pattern (and the choice of sentence-final particles) associated with the disapproval construal. I will leave the experimental account to future research.

without changing their semantics, as evidenced by (6) and (7) respectively:³

- (6) 阿 Q 怎麼 會 老是 遲到? (!) (causal-mirative *how*)
 Akiu **zenme** **hui** laoshi chidao?(!)
 Akiu how will always late
 ‘How come Akiu was always late?(!)’
- (7) 怎麼 阿 Q 會 這樣 對 你! (denial-disapproval *how*)
Zenme Akiu **hui** zheyang dui ni!
 how Akiu will this.manner treat you
 ‘How can Akiu treat you like this/as such!’

Likewise, *zenme* ‘how’ merges low to form a typical instrumental question such as (8), which again can be paraphrased with the modal *hui*, as in (9):

- (8) 阿 Q 平常 怎麼 燒 這道 菜? (instrumental *how*)
 Akiu pingchang **zenme** shao zhedao cai?
 Akiu usually how cook this.CL dish
 ‘Usually how did Akiu cook this dish?’
- (9) 阿 Q 平常 會 怎麼 燒 這道 菜? (instrumental *how*)
 Akiu pingchang **hui** **zenme** shao zhedao cai?
 Akiu usually will how cook this.CL dish
 ‘Usually how did Akiu cook this dish?’

Furthermore, the strategy of combining a *wh* and a modal in forming a non-canonical *wh*-question is quite productive in Classical Chinese: As observed in Lau & Tsai (2020), *an* is typically interpreted as ‘where’ in a locative *wh*-question like (10). By contrast, when *an* appears in front of a modal such as *neng* ‘can,’ as in (11), its reading is very much in line with the denial-disapproval usage of *zenme*, forming a non-canonical *wh*-question. It is in this respect that Classical Chinese and English seem to pattern together in typological terms.

- (10) 沛公 安 在?
 Peigong **an** zai?
 Duke.of.Pei where at
 ‘Where is Duke of Pei?’
 (Excerpt from *Records of the Grand Historian—Biography of Xiang Yu* [史記・項羽列傳])
- (11) 暴 而 不 戢, 安 能 保 大?
 Bao er bu ji, **an** neng bao da?
 violent and not stop how can remain strong
 ‘If a state is violent and does not refrain from waging wars, how can it remain strong?’
 (Excerpt from *Zuo Zhuan—Duke Xuan Year 12* [左傳・宣公十二年])

On the other hand, modern Chinese differs from this pattern in turning on the silent mode of the modal component in licensing non-canonical *wh*-construal, presumably due to the prominence of its mood and modal categories over tenses (Mei 2015, Tsai 2019). As a matter

³ The at-issue proposition of (4) would be that Akiu was always late for some cause, while the not-at-issue content states that Akiu’s being late is unexpected. At this stage, the information-seeking function is still there. By contrast, the full-fledged force shift applies to (5) in that the speaker is no longer interested in the answer to the causal question (i.e., the original at-issue content), and the speech act is distinctly disapproval rather than interrogative.

of fact, we can still spot this trait with the obligatory presence of *ē* ‘will’ and *thang* ‘can’ in the following Taiwan Southern Min (TSM) non-canonical *wh*-questions:

- (12) 伊 哪 *(會) 無 來? !
 I ná *(ē) bô lâi?!
 s/he how will not come
 ‘How come s/he wouldn’t come?!’ or ‘Why the hell wouldn’t s/he come?!’
- (13) 汝 哪 *(通) 做 彼 款 代誌? !
 Lí ná *(thang) tsò hit khuán tāitsì?!
 you how can do that kind thing
 ‘How could you do such a thing?!’

Our analysis thus introduces another strategy of forming non-canonical *wh*-questions: First note that there is a curious tone sandhi pattern of (*sī*)-*án*-*tsuánn* ‘(Foc.)how’ in TSM, as illustrated by the contrasts between (14)–(15) and (16) (see also Lau & Tsai 2020):

- (14) 是 按 怎 水 順 咧 無 歡 喜 啊? ! (denial-disapproval *how*)
Sī-án-tsuánn⁵¹ Tsuí-sūn teh bô huann-hí ah?!
 FOC.how Tsuisun ASP not happy SFP
 ‘How the heck is Tsuisun being unhappy! (He shouldn’t be.)’
- (15) 水 順 按 怎 咧 無 歡 喜 啊? ! (causal *how*)
 Tsuí-sūn **án-tsuánn**⁵¹ teh bô huann-hí ah?!
 Tsuisun how ASP not happy SFP
 ‘How come Tsuisun is being unhappy?! (This is not expected.)’ (Lau & Tsai 2020:262)
- (16) 水 順 按 怎 咧 無 歡 喜 啊? (manner *how*)
 Tsuí-sūn **án-tsuánn**⁵⁵ teh bô huann-hí ah?
 Tsuisun how ASP not happy SFP
 ‘In what way did Tsuisun show his unhappiness?’ (Lau & Tsai 2020:262)

The denial-disapproval *how* in (14) and the causal *how* in (15) are pronounced with a falling tone (i.e., its citation tone 51), whereas the manner *how* in (16) carries a high-level tone (i.e., the sandhi tone 55). In addition, the denial-disapproval *how* comes with a focus marker (i.e., *sī* prefixed to *án-tsuánn* ‘how’ with prominent stress). This prosody-semantics correspondence thus lends support to the high-low dichotomy in question: Namely, denial-disapproval *how* and causal *how* merge high to form an independent tone sandhi domain in the left periphery, while manner *how* merges low to form a tone sandhi domain with *vP* (i.e., the main predicate). As a result, it is not unreasonable to assume that, just like in Mandarin, there is an implicit modal in TSM that follows outer *how* in (14)–(15) and precedes inner *how* in (16). We therefore propose that it functions as a delimitator of the two distinct tone sandhi domains in prosodic terms.

Further evidence can be drawn from the combo of *dim*² ‘how’ and *wui*⁵ ‘will’ in Cantonese, as in (17a) and (18a):

- (17) a. 佢 點 會 走? !
 Keoi⁵ dim² wui⁵ zau²?!
 he how will leave
 ‘How can he leave?!’
- b. *佢 點 走? !
 *Keoi⁵ dim² zau²?!
 he how leave

- (18) a. 佢 點 會 咁 叻? !
 Keoi² dim² wui⁵ gam³ lek¹?!
 he how will so good
 ‘How can he be so good?!’
 b. *佢 點 咁 叻? !
 *Keoi² dim² gam³ lek¹?!
 he how so good (Tang 2009:241)

Just like *ē* and *thang* in TSM, the modal component cannot be omitted, as evidenced by (17b) and (18b) respectively (cf. Kwok 2003, Tang 2009).

3. Prosody matters

Another aspect of non-canonical *wh*-construals stems from the prosodic arrangement of the entire sentence or certain focused areas. In addition to the usual factors such as intonation and clause-typing sentence final particles, we find a peculiar construction where stress placement and silent categories are also involved, namely, the so-called whining *what*-question. As observed by Tsai (2011, 2021), it differs from a typical question like (19a) in involving an implicit light verb FOR. In other words, it is essentially a reason/purpose applicative in disguise with an overriding whining force and accompanying exclamatory intonation, as in (19b).

- (19) a. 你 看 什麼? (interrogative *what*)
 Ni kan shenme?
 you look.at what
 ‘What are you looking at?’
 b. 你 看 什麼? ! (whining *what*)
 Ni kan shenme?!
 you look.at what
 ‘What the heck are you looking at (me) for?!’

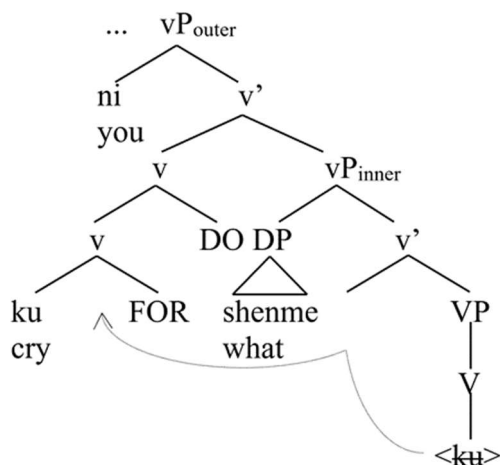
Also, it is noteworthy that only the whining construal may involve verb copying at the end of the sentence, as evidenced by the contrast between (20a) and (20b) (Q_{wh} : *wh*-question particle; Q_{ex} : exclamative particle):

- (20) a. *你 看 什麼 看 (呢)? (interrogative *what*)
 *Ni kan shenme kan (ne)?
 you look.at what look.at Q_{wh}
 b. 你 看 什麼 看 (啊)? ! (whining *what*)
 Ni kan shenme kan (a)?!
 you look.at what look.at Q_{ex}
 ‘What the heck are you looking at (me) for?!’

Now consider a simpler example, the unergative construction (21a): it involves a silent inner light verb FOR that attracts the main verb *ku* ‘cry’ (cf. Tsai 2011), and then raises as a cluster to the outer light verb DO (an eventuality predicate in Huang’s 1994, 1997 terms), forming the apparent VO order on the surface, as sketched in the tree diagram (21b).

- (21) a. 你 哭 什麼 (哭)? !
 Ni ku shenme (ku)?!

- you cry what cry
 ‘What the heck are you crying for?! (You shouldn’t be crying.)’
 b.



(Tsai 2021:208)

This particular analysis enjoys the advantage of killing two birds with one stone: That is, the verb copying option receives a natural account once we adopt the copy theory of head movement (cf. Chomsky 1995), where the verb copy left behind may or may not delete due to the focus property and prosodic prominence carried by FOR (see below). Moreover, it explains away the puzzle why an unergative verb like *ku* ‘cry’ may have a direct object, as well as the fact that the sentence is essentially an applicative *why*-question in disguise (cf. Tsai 2021).

But where does the whining force come from? How is it encoded in Mandarin? As demonstrated by the experimental study of Yang & Tsai (2019), the force shift from interrogative to whining has a lot to do with the change of the overall prosodic pattern (i.e., the distinctive intonation associated with clause-typing), as well as the stress shift from the object *wh*-expression (i.e., the locus of nuclear stress) to the inner light verb FOR, which is the locus of focus (also cf. Tsai 2011, 2021). Most importantly, when the inner light verb is silent, it needs something to carry the prosodic weight assigned to it. This in turn triggers the V-to-FOR raising in (10b). Verb copying associated with similar focus effects are also observed in Gungbe (Aboh 2004) and the Kwa languages (Landau 2007).

In terms of technical treatments, force shift is done on a par with the licensing of *wh*-expressions through unselective binding, except that the Q-operator has been replaced by a whining operator in the left periphery (cf. Lau & Tsai 2020, Tsai 2021). As a result, the illocutionary force has shifted from interrogative to whining and/or disapproval.

Take (19a–b) for instance: As illustrated in Figure 1, the pitch arises at the end of a canonical *what*-question, but not for the intonation of its non-canonical counterpart.⁴

⁴ Figures 1 through 5 are adapted from Yang & Tsai (2019). It is noted there that the experiment was done based on the tests on 18 people.

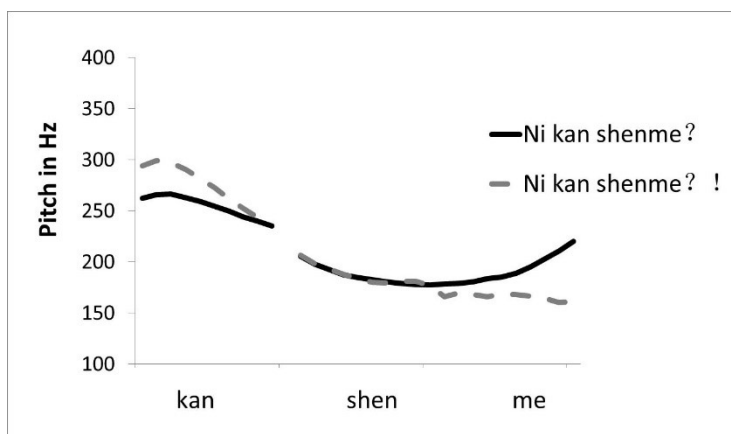


Figure 1. Mean F0 contour of whining *wh* (dotted line) vs. interrogative *wh* (solid line) (Yang & Tsai 2019:41)

Next, the figure below shows that the duration of the interrogative *shenme* ‘what’ is much longer than its whining counterpart, presumably due to the information focus carried by the former.

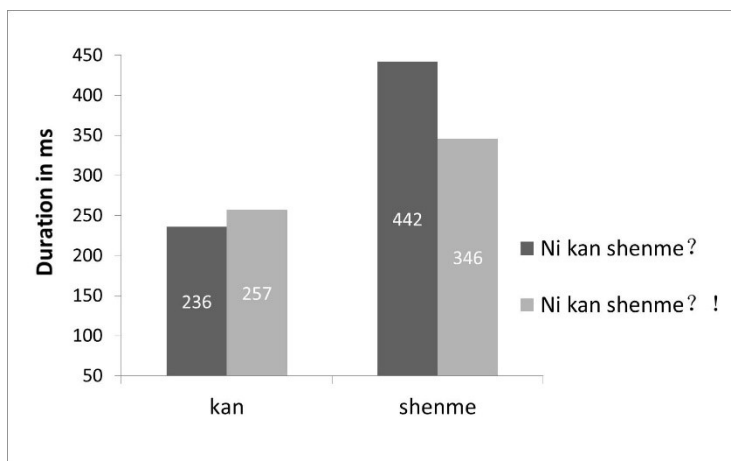


Figure 2. Mean duration of whining *wh* (grey) vs. interrogative *wh* (black) (Yang & Tsai 2019:41)⁵

Finally, it is demonstrated in Figure 3 that the intensity of the verb in the whining sentence is much greater than its counterpart in the interrogative sentence.

⁵ There is an inconsistency between the duration measurement of *shenme* ‘what’ as a unit in Figure 2 and the separation of *shen* and *me* with respect to the intensity measurement in Figure 3. Fortunately, this does not affect our overall observation that interrogative *shenme* in (19a) is much more prominent than its whining counterpart in (19b) in prosodic terms.

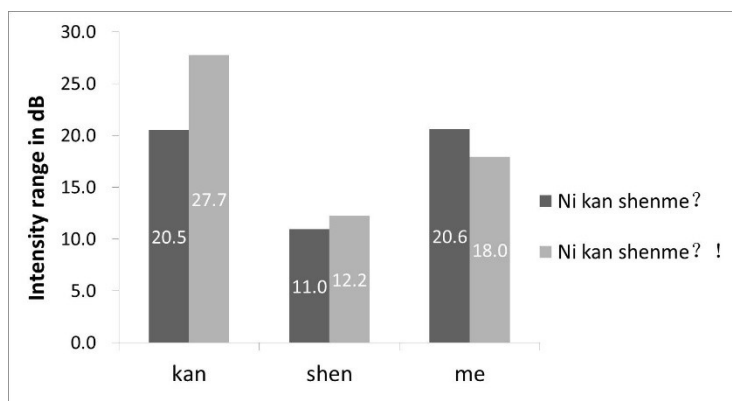


Figure 3. Mean intensity range of whining *wh* (grey) vs. interrogative *wh* (black) (Yang & Tsai 2019:41)

Interestingly enough, when we examine the construction where verb copying does apply, i.e., in the presence of whining *wh*-construal, the pitch of the first verb copy is consistently higher than that of the second one, as sketched in the following figure.

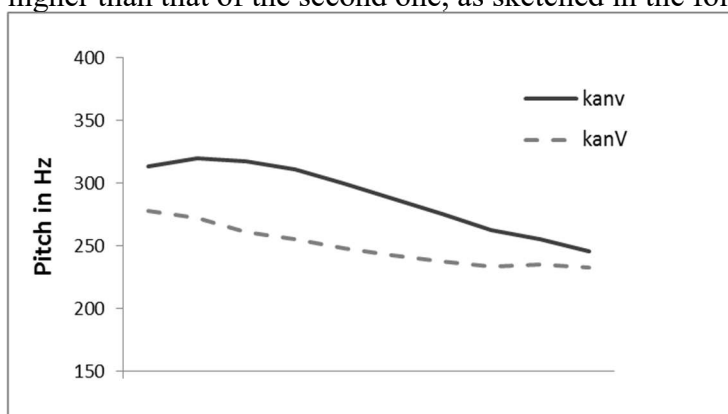


Figure 4. Mean F0 curves of the two verb copies (Yang & Tsai 2019:42)

Furthermore, Figure 5 shows that the intensity of the first copy is much stronger than that of the second copy.

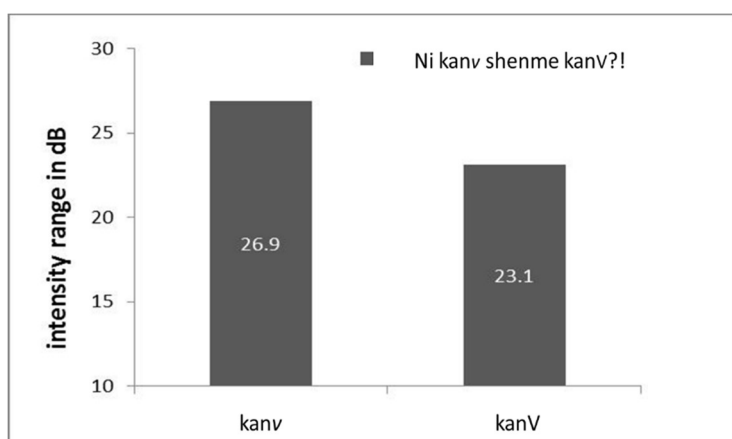


Figure 5. Mean intensity range of the two verb copies (Yang & Tsai 2019:42)

For one thing, the second copy of the verb chain in (21b) is subject to PF deletion. This is because spelling out both copies violates the Linear Correspondence Axiom (LCA) (cf. Kayne 1994). For another, according to Cheng (2007), verb copying applies at failure to reduce a verb

chain, as its lower copy has been fused with an aspect marker. In our case, raising to FOR is more in line with the raising to Foc along the line of Hornstein & Nunes (2002) and Nunes (2004): As proposed by Tsai (2021), it triggers a morphological fusion between the main verb and the light verb, even if the latter category is silent in Mandarin. The experimental study presented above thus provides direct evidence for our claim that the verb copying of (20b) and (21a) is made possible by the focus feature associated with the light verb FOR (as indicated by the prosodic prominence of the first verb copy).

Further cross-dialectal support can be drawn from the following whining *what*-question in Taiwan Southern Min (cf. Lau & Tsai 2020).

- (22) 汝 是 咧 哭 啥? !
 Lí sī leh khàu siannh?!
 you FOC LEH cry what
 ‘What the heck are you crying for?! (You shouldn’t be crying.)’ (Lau & Tsai 2020: 259)

Here *sī leh* serves as a complex verum focus marker in the preverbal position, which in turn assigns prosodic prominence to the main verb *khàu* ‘cry.’ Therefore, it can be regarded as a lexical counterpart of the focus feature associated with the silent light verb FOR in Mandarin whining *wh*-construals.⁶

On the cross-linguistic front, Phan & Tsai (2022) explore the extensive usage of Vietnamese whining questions, as seen in (23), which sports exactly the same verb copying pattern as their Mandarin counterparts with a rising-falling intonation:

- (23) Mày khóc gì mà khóc?!
 2SG cry what PRT cry
 ‘What are you crying for?!’ (Phan & Tsai 2022:169)

As pointed out by Vanrell (2013), this prosodic pattern very often encodes a particular type of the speaker’s attitude, i.e., disapproval, surprise or irritation.

In addition, as noted by Ishida, Hsieh & Tsai (2024), the distinct accent placement in (25) contrasts with (24). Here *nani* ‘what’ can be construed as whining instead of interrogative in Japanese (the capital characters representing the loci of accent).

- (24) NAni-o NOnderu no? (interrogative *what*)
 what-ACC drinking Q
 ‘What are you drinking?’
 (25) NaNI-O NOnderu no?! (whining *what*)
 what-ACC drinking Q
 ‘What the hell are you drinking for! (Stop drinking!)’

As illustrated in Figure 6, the intonation of the whining sentence (dotted line) starts much lower than that of its interrogative counterpart (solid line) but finishes flat without rising like that of (24).⁷

⁶ A verum focus results from accentuation of a specific component in the left periphery, emphasizing the truth of a proposition (cf. Lohnstein 2018). Special thanks to an anonymous reviewer for raising the issue concerning the placement of focus in relation to prosodic prominence.

⁷ Figure 6 is adapted from a pilot study in Ishida, Hsieh & Tsai (2024), which is based on the acoustic data from three Kanto Japanese speakers. This ongoing project will eventually involve another 10 informants to give a more comprehensive analysis on the prosodic pattern in question.

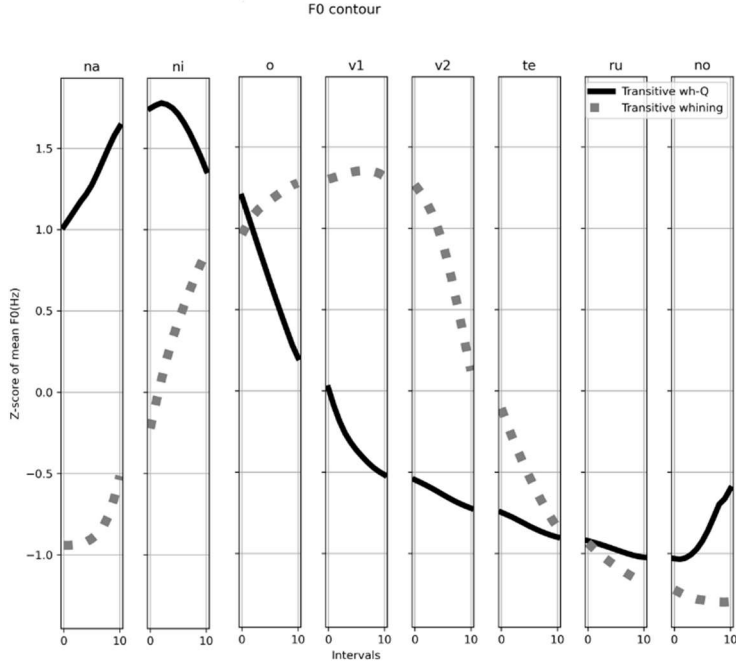


Figure 6. Mean F0 contour of *nani*-questions in Japanese (Ishida, Hsieh & Tsai 2024:22)

4. Negative modality in action

It is also important to note that Mandarin whining *what* has a counterpart in the left periphery, as in (26a), which carries the disapproval/denial force (cf. Shao 1996, Cheung 2008, Tsai 2011, Pan 2015, Yang 2021, among others). As pointed out by Tsai (2015b:292), this type of non-canonical *shenme* functions like a negative modal; it serves as a sentential adverbial, and triggers negation over epistemic modality. By contrast, the whining *shenme* in (26b) merges low in the verbal domain, and the reading is distinctively deontic under negation (hence LWHAT vs. HWHAT in Yang’s terms).

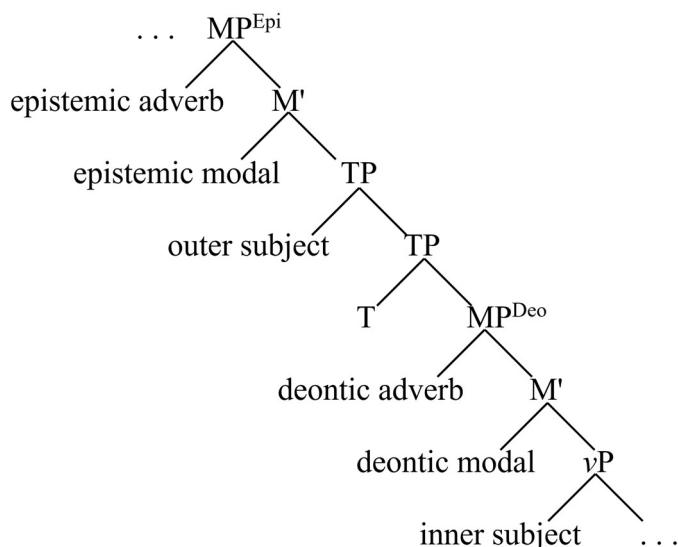
- (26)a. 什麼 阿Q 去了 北京！ 他 去了 上海。
Shenme Akiu qu-le Beijing! Ta qu-le Shanghai.
 what Akiu go-PRF Beijing he go-PRF Shanghai
 ‘Akiu couldn’t have gone to Beijing! He has gone to Shanghai.’
 b. 阿Q 去 什麼 北京！ 他 這裡 工作 都 沒 做完。
 Akiu qu **shenme** Beijing! Ta zheli gongzuo dou mei zuowan.
 Akiu go what Beijing he here work all have.not finish
 ‘Akiu shouldn’t go to Beijing! He hasn’t even finished the work here.’

In other words, there is an epistemic vs. deontic distinction corresponding to the “height of interpretation” of non-canonical *what*-questions in Mandarin.

There are two ways to approach this issue: one is to attribute the negative modal construals to pragmatic implicatures under force shift. However, it is unclear how this approach may drive the high-low dichotomy illustrated above. The other is to resort to the silent modal analysis given that the locus of an epistemic modal is typically associated with the left periphery, whereas that of a deontic modal is situated in the neighborhood of *vP* (cf. Butler 2003, among others), with the former responsible for the non-canonical reading of (26a), and the latter for that of (26b). As it turns out, the epistemic-deontic distinction corresponds perfectly to the topography of Mandarin modals laid out in Tsai (2015b), as illustrated in the following

diagram:⁸

(27)



(Tsai 2015b:288)

We will therefore pursue the latter option, putting forth the claim that the *wh* component shifts its force from interrogative to denial/disapproval through mirativity, in that the modal component provides the epistemic state against which the speaker evaluates the not-at-issue content, i.e., his or her expectation (cf. Rett 2012). This move, once verified, lends further support to our claim that a non-canonical *wh*-question should be decomposed into two parts, i.e., *wh* + modal, even if the latter can be silent due to typological settings.

From a comparative syntactic viewpoint, we may adopt Tsai's (2019) view that Mandarin is a mood-prominent language, where modals may remain silent at PF as one of its major typological features. The distribution of these silent modals, just like their lexical counterparts, observes the epistemic-deontic split along the clausal spine, i.e., the former corresponding to the CP phase, and the latter the vP phase (cf. Butler 2003, Tsai 2015b). This lays the ground

⁸ An anonymous reviewer points out that it is not immediately clear how the proposed silent modal analysis works for negative scopal interaction in (26b). As noted by Iatridou & Zeijlstra (2013), English universal deontic modals can be divided into two groups in terms of their interaction with negation: *should* and *must* take wide scope, as in (ia), whereas *have to* and *need to* take narrow scope, as in (ib):

(i) a. He should/must not go.

b. He doesn't have/need to go.

Interestingly, Mandarin deontic modal such as *yinggai* 'should' can work both ways, as evidenced by (iia) in contrast to (iib):

(ii) a. 他 應該 不 去。

Ta yinggai bu qu.

he should not go

'He should not go.'

b. 他 不 應該 去。

Ta bu yinggai qu.

he not should go

'He is not supposed to go.'

We therefore propose that its silent counterpart works exactly the same way when construed with non-canonical *what* in (26b), which does allow negation scoping over the deontic modal. Besides, it is highly possible that the silent deontic modal belongs to the *have to* group, and gets interpreted as 'be supposed to.'

Furthermore, it follows from the mood-prominent nature of Mandarin that the temporal interpretation of a bare sentence like (26b) can be either realis or irrealis (cf. Mei 2015, Tsai 2019). The mood category may well reside in the head position of TP in (27) (or that of FinP in Rizzi's 1997 sense).

for the distinct modal bases for (26a) vs. (26b). On the other hand, non-canonical *shenme* ‘what’ triggers negation through the syntax-pragmatics interface by disregarding its information-seeking function in favor of negative attitude/force. This may well be the origin of negation over the silent modals in question.

As a matter of fact, we can find traces of silent modals even in Classical Chinese. Hu (2023) points out that *he* ‘what’ in the following *The Analects* text actually involves implicit modality, which, according to his analysis, raises to the CP domain to substantiate the construal:

- (28) 出 則 事 公卿； 入 則 事 父兄；
 Chu ze shi gongqing; ru ze shi fuxiong;
 go.out then serve minister enter then serve father.brother
 喪事 不 敢 不 勉；
 sanshi bu gan bu mian;
 funeral NEG dare NEG diligent
 不 為 酒 困—— 何 有 於 我 哉！
 bu wei jiu kun—— he you yu wo zai!
 NEG by alcohol troubled what have toward I SFP
 ‘When abroad, I serve the ministers and nobles; at home, I serve my elders;
 for funeral matters, I do not dare to be negligent; and I do not let myself become
 impaired by alcohol –What more CAN be asked of me!’
 (Excerpt from *The Analects*—*Zi Han* [論語・子罕])

Along our line of reasoning, this means there is an implicit possibility modal employed in conjunction with the *wh*-word *he*, both high up in the left periphery. The combination thus triggers the force shift from interrogative to exclamative.

Another suggestive observation is made by Meisterernst (2020, 2025): Classical Chinese silent modals may well appear under the negation, which can occupy two distinct syntactic positions, one in the CP layer and the other in the IP domain. Interestingly, the former often triggers a rhetorical question such as (29), while the latter expresses prohibition (i.e., negative deontic modality), as in (30).

- (29) 毋 乃 不 可 乎？！
 Wú nǎi bù kě hū?!
 NEG_{mod} then NEG possible Q
 ‘Shouldn’t that be inappropriate?!’
 (Excerpt from *Zuo Zhuan*—*Duke Xuan Year 15* [左傳・宣公十五年])
- (30) 王 必 勿 與。
 Wáng bì wù yǔ.
 king certainly NEG_{mod} give
 ‘Your majesty certainly should not give it.’
 (Excerpt from *Strategies of the Warring States* [戰國策])

5. In the grand scheme of things

Finally, let’s take a step back, and consider the origin of causal/mirative questions such as (31) and (32) on conceptual grounds.

(31) **How come** you talk like that?!

(32) 你 怎麼 那樣 說話？！

Ni **zenme** nayang shuohua?!
you how that.manner talk
'How come you talked in such a manner?!'

Both *how come* and causal *zenme* ‘how’ questions how the talking event comes about with a distinct flavor of unexpectedness (hence mirativity). Tsai (2008, 2015a) suggested that they are hosted by a cause projection (CauP) in the left periphery, since the construals presuppose an external cause. This interpretation presents a sharp contrast with an epistemic *why*-question, as shown by the semantic/pragmatic difference between (33) and (34):

- (33) **How come** the sky is blue?!
- ≈ The sky shouldn't be blue, and there must be something causing this phenomenon.
- (34) **Why** is the sky blue?
- ≈ Given that the sky is blue, what would be (scientific) reason for it?

Exactly the same contrast can be found with their Mandarin counterparts in (35) and (36):

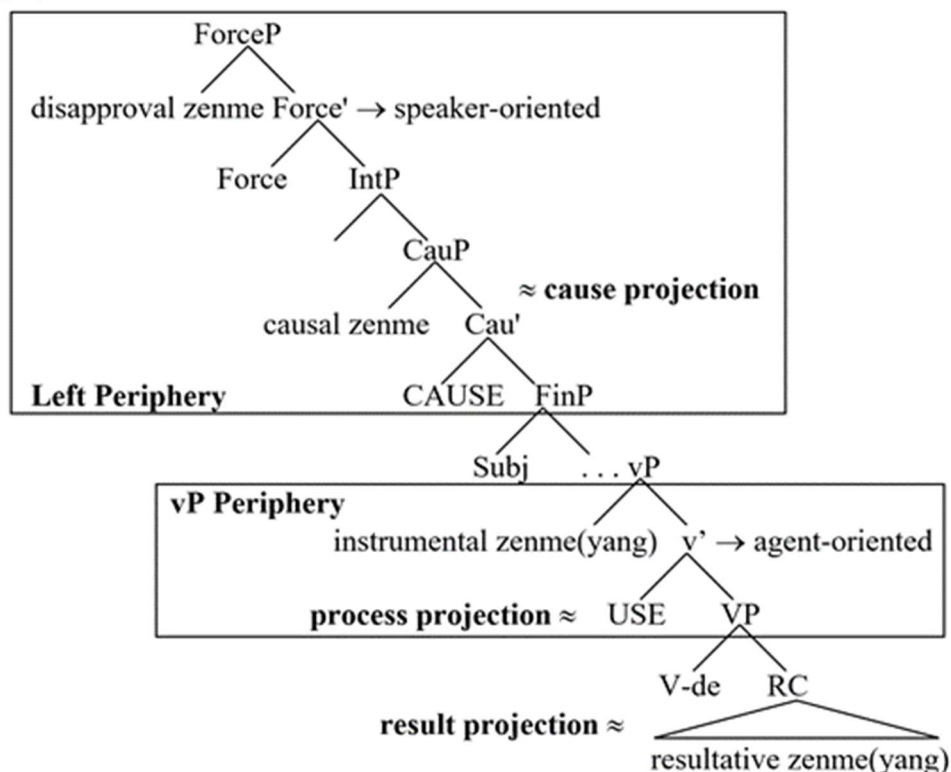
- | | | | | | | |
|------|---|-----------|-----|------|--------------------------|--|
| (35) | 天空 | 怎麼 | 是 | 藍 | 的？！ | |
| | Tiankong | zenme | shi | lan | de?! (causal question) | |
| | sky | how | be | blue | DE | |
| | 'How come the sky is blue?! (The sky shouldn't be blue.)' | | | | | |
| (36) | 天空 | 爲什麼 | 是 | 藍 | 的？ | |
| | Tiankong | weishenme | shi | lan | de? (epistemic question) | |
| | sky | why | be | blue | DE | |
| | 'Why is the sky blue?' | | | | | |

To substantiate this interpretive distinction in syntactic terms, we follow Shlonsky & Soare (2011) in placing *why* (and presumably *weisheme*) in a reason projection (ReasonP) in the left periphery.

Also, it is worthwhile to note that only the causal question, but not the epistemic question, gives rise to the mirative construal of (33), as well as that of its Mandarin counterpart in (35). We would like to argue that this is exactly where a causal construal departs from its interrogative origin: Since the speaker cannot accept the at-issue content, they are no longer interested in the answer. It follows that the force shift applies when the causal-effect relationship is disrupted, presumably at the top of the clausal spine. This point is best illustrated by the causality hierarchy of *zenme(yang)* 'how(manner),' as shown in the following topography of its distribution:⁹

⁹ See Tsai (2024) on how this cartographic arrangement results from extending the cause-process-result hierarchy in Ramchand's (2008) sense from the first phase (i.e., the vP domain) to the second phase (i.e., the CP domain).

(37)



Along this line, we may further extend the above account further to the non-canonical usage of *where*-questions like (38), through which the speaker raises an objection to the addressee's previous remarks:¹⁰

(38) Where does that come from?! (You are being unreasonable!)

Another nice demonstration of this construal comes from *nar* 'where' in the following Mandarin example, which is actually ambiguous between (39a) and (39b) (cf. Tsai 2011):

(39) 您 這 是 哪兒 的 話 ？ ！

Nin zhe shi nar de hua?!

you this be where DE dialect

a. Lit. 'Where's dialect is this (speech) of yours?'

(*where*-question)

b. 'Don't mention it!' (polite dismissal)

(Tsai 2011:198)

Even more interestingly, the locative expression *nali* 'where' has evolved into a fully grammaticalized negator to mark out the disagreement among interlocutors (see also Cheung 2008), as exemplified below:

(40) 他 哪裡 去了 台北 ！

Ta nali qu-le Taipei!

he where go-PRF Taipei

'There is no way he went to Taipei!'

¹⁰ For various intriguing denial/refutatory usages of *đâu* 'where' in Vietnamese, as well as a fine-grained cartographic account, see Phan & Tsai (2025).

Here we propose that non-canonical *where*-expressions should not be construed as locative. Rather, they involve a source-goal relation, which is on a par with the cause-effect relation encoded by *how come* and causal *zenme*. On the same conceptual ground, we may attribute the force shift in (39) and (40) to the disruption of a causal/source question, which in turn gives rise to a disapproval/refutatory force and/or a negative attitude. Along this line, we may well put the hierarchy of (41a–b) side by side as the origin of non-canonical construals of *how*, *where* and *for what* across languages:

- (41) a. Cause > Process > Effect
 b. Source > Process > Goal

In terms of morphosyntax, suggestive evidence can also be drawn from the causal usage of locative expressions across languages. For instance, *therefore* is interpreted as ‘because of that,’ while *wherefore* is interpreted as ‘because of what.’ On the other hand, the original meaning of *hence* is ‘from here,’ whereas that of *whence* is ‘from where.’ Both eventually derive the causal usage.

6. Concluding remarks

In conclusion, although we do not attempt to provide a full-fledged semantic account of the negative modality involved in non-canonical *wh*-construals, it is likely that they are encoded by various functional elements along the clausal spine, as well as by prosodic arrangements such as stress and intonation. We therefore conclude that a non-canonical *wh*-construal may contain the following four components:

- (42) a. On the morpho-syntactic front, it combines an applicative/oblique *wh*-expression and a modal (which can be silent), very often accompanied by a focus/mood marker.
 b. On the prosodic front, it is associated either with a peculiar intonation, or with a stress shift onto certain functional elements.
 c. On the semantic front, it involves negation over modal quantification in relation to not-at-issue content such as expectations and/or presuppositions.
 d. On the pragmatic front, it substantiates a non-interrogative speech act, raising objection to either the at-issue content or the remarks/behavior of an interlocutor. We also explore the conceptual connection between the causality hierarchy and the source hierarchy to pinpoint the origin of non-canonical *wh*-construals.

All in all, this pilot study serves only to reveal the tip of an iceberg as to how non-canonical questions are constructed across languages. There are plenty of important issues to be solved in the future, including but not limited to the origin of negative modality, the interface properties of causation and causality, as well as the intriguing relationship between grammaticalization and the mechanism of force shift.

List of abbreviations

ACC	accusative case marker
ASP	aspectual marker
BA	affective light verb
CL	classifier
DE	modifier marker
FOC	focus marker

NEG	negator
NEG _{mod}	modal negator
PRF	perfective aspect marker
PRT	particle
SFP	sentence-final particle
Q	question particle
Q _{ex}	exclamative particle
Q _{wh}	<i>wh</i> -question particle

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